

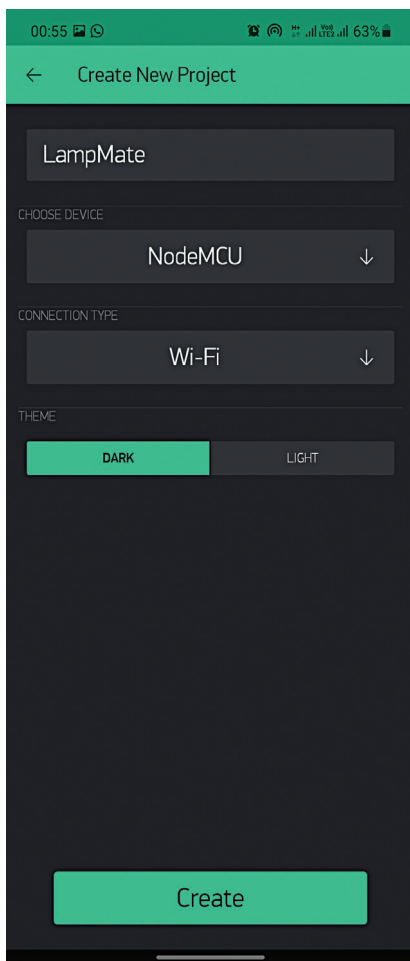


Fig. 8: Creating New Project window

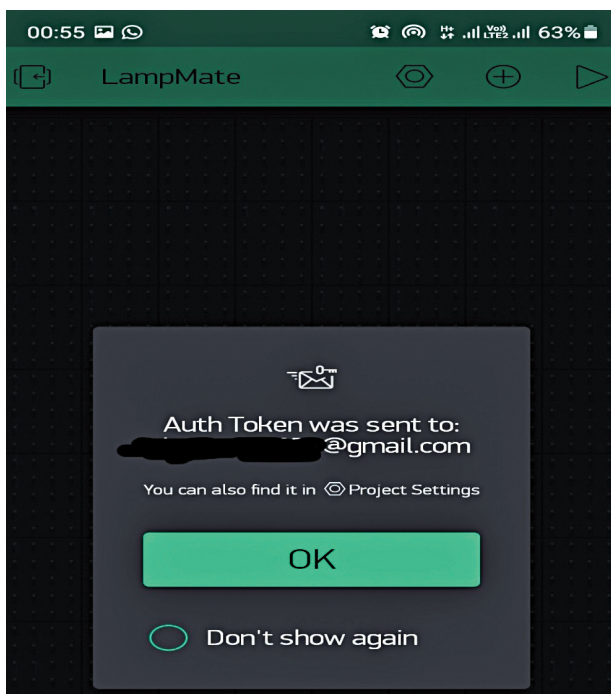


Fig. 9: Authorisation token notification window

physical board that stores and executes the program code uploaded to it. The software package and the board together are referred to as Arduino.

Blynk. Blynk IoT platform offers a full suite of software that allows to prototype, deploy, and remotely manage connected electronic devices at any scale, from small IoT projects to millions of commercial connected products.

Hardware and software setup

Connect NodeMCU to your PC. Open the Arduino code *transmitter_IR.ino* from Arduino IDE and select the correct board and COM port from Arduino IDE. Select the board as NodeMCU 1.0 (ESP-12E module) as shown in Fig. 3. The author's IR sensor transmitter fixed on the chair is shown in Fig. 4 and author's receiver circuit wired on breadboard is shown in Fig. 5.

Fig. 5.

Before proceeding further, make sure you have the following libraries installed in your Arduino IDE:

- Blynk library
- Blynk simple library
- ESP8266 Wi-Fi library

Before uploading the code *transmitter_IR.ino* to Board1, you need to include your own SSID, password, authorisation token in the code and compile it. Similarly, include these details in the receiver code *receiver_lamp.ino* in Board2. The authorisation

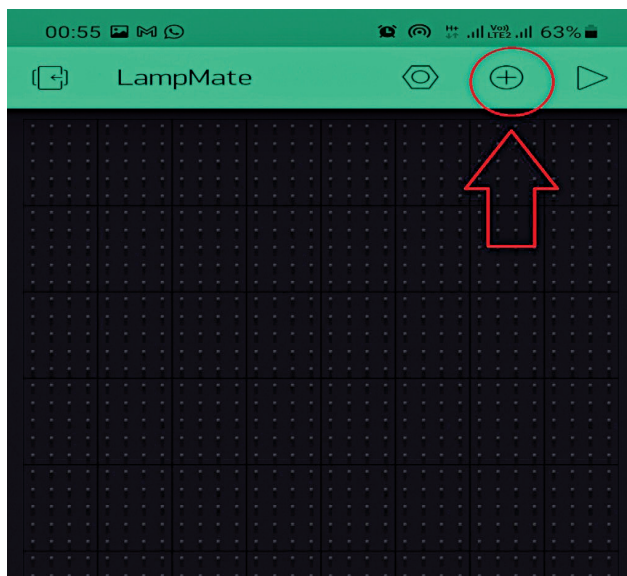


Fig. 10: Clicking on + (plus) icon

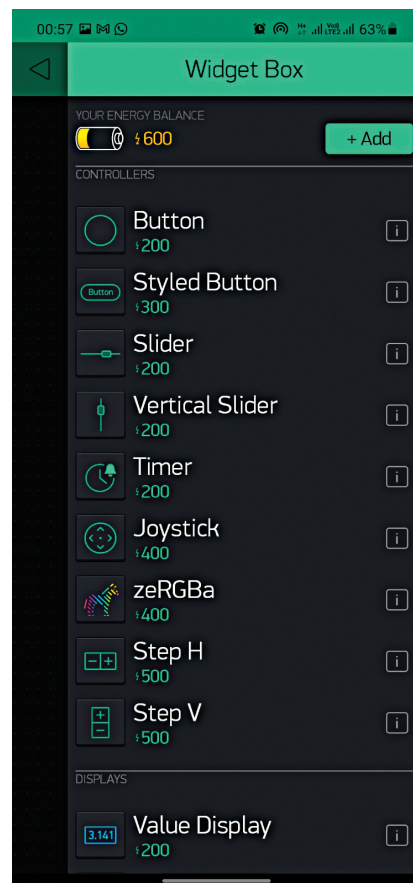


Fig. 11: Widget box

token is generated from Blynk app, which is explained below.

Blynk app configuration

1. Download the Blynk app in your smartphone from Play Store or from